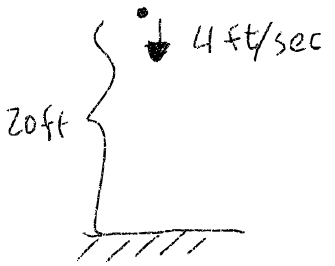
CHAPTER 5



• $20 = -4t - \frac{1}{2}(9.8)t^2$

$4.9t^2 + 4t - 20 = 0$

$t = \frac{-4 \pm \sqrt{16 + 80(4.9)}}{9.8}$

$\approx \frac{-4 \pm 20.2}{9.8}$

≈ 1.653 sec

$a(t) = -32$ ft/s^2

$v(t) = -32t + c$

$v(0) = -32 + c = -4$

$c = 28$

$v(t) = -32t + 28$

$y(t) = -16t^2 + 28t + D$

$y(0) = D = 20$

Exercise 5.1.a $\int (5x^4 - 8x^3)\,dx = x^5 - 2x^4 + C$

(5.1.b) $\int \frac{1}{\sqrt{1-x^2}}\,dx = \sin^{-1}(x) + C$

(5.1.c) $\int \cos(5x)\,dx = \frac{1}{5}\sin(5x) + C$

(5.1.d) $\int \frac{1}{x}\,dx = \ln x + C$

(5.1.e) $\int \frac{1}{5+x}\,dx = \ln(5+x) + C$

(5.1.f) $\int \frac{5}{1+z^2}\,dz = 5\tan^{-1}(z) + C$

(5.1.g) $\int (x + 2 - x^{-2})\,dx = \frac{1}{2}x^2 + 2x + \frac{1}{x} + C$

(5.1.h) $\int \left(x^{3/2} + x^{-11/6}\right)dx = \frac{2}{5}x^{5/2} + \frac{6}{5}x^{5/6} + C$

(5.1.i) $\int e^{-2t}\,dt = -\frac{1}{2}e^{-2t} + C$

(5.1.j) $\int \frac{1}{3-2x}\,dx = -\frac{1}{2}\ln(3-2x) + C$

Exercise 5·2

(a) $\int \underbrace{\cos x}_{\text{height}} \underbrace{dx}_{\text{width}} = \sin x + C$

(b) $\int_0^1 2x \cdot dx = \left[x^2\right]_0^1 = 1^2 - 0^2 = 1$

(c) $\int_0^1 \frac{1}{1+t^2}\, dt = \left[\tan^{-1}(t)\right]_0^1 = \tan^{-1}(1) - \tan^{-1}(0)$

$= \frac{\pi}{4}$

Exercise 5·3

$f'(x) = 3x^2 - 3x + 1$

(a) no we need "initial condition"

$f(x) = \int f'(x)\, dx = x^3 - \frac{3}{2}x^2 + x + C$

(b) $f(1) = 1 - \frac{3}{2} + 1 + C = 2$

$C = \frac{3}{2}$

$f(x) = x^3 - \frac{3}{2}x^2 + x + 2$

(c)



The candidate antiderivatives differ from each other by an additive constant. So, they have exact same shape just at different heights.

value at one x gives area, the height thus a unique antideri

Exercise 5.4

(a) $f'(x) = \sqrt{x}\left(1 + \frac{5x}{2}\right)$

$$f(x) = \int f'(x)\,dx = \int \left(\sqrt{x} + \frac{5}{2}x^{3/2}\right)dx$$

$$= \frac{2}{3}x^{3/2} + x^{5/2} + C$$

$$f(4) = \frac{2}{3}\cdot 8 + 32 + C = 32$$

$$C = -\frac{16}{3}$$

$$f(x) = \frac{2}{3}x^{\frac{3}{2}} + x^{\frac{5}{2}} - \frac{16}{3}$$

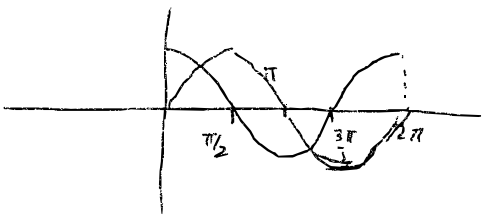
(b) $f''(x) = \sin x - \cos x$

$$f'(x) = \int f''(x)\,dx = -\cos x - \sin x + C$$

$$f'(0) = -1 - 0 + C = 3$$

$$C = 4$$

$$f'(x) = -\cos x - \sin x + 4$$



$$f(x) = \int f'(x)\,dx = -\sin x + \cos x + 4x + D$$

$$f\left(\frac{3\pi}{2}\right) = -\sin\left(\frac{3\pi}{2}\right) + \cos\left(\frac{3\pi}{2}\right) + 6\pi + D = 1$$

($\sin\frac{3\pi}{2} = -1$, $\cos\frac{3\pi}{2} = 0$)

$$D = -6\pi$$

$$f(x) = \cos x - \sin x + 4x - 6\pi$$

Exercise 5.4.e

$f''(x) = 4 + 6x$

$f'(x) = \int f''(x)\,dx = 4x + 3x^2 + \alpha$

$f(x) = \int f(x)\,dx = 2x^2 + x^3 + \alpha x + \beta$

$f(1) = 2 + 1 + \alpha + \beta = 1$; $\boxed{\alpha + \beta = -2}$

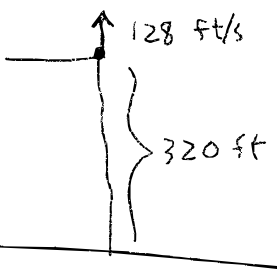
$f(-1) = 0 = 2 - 1 - \alpha + \beta$; $\boxed{\alpha - \beta = 1}$

$\alpha = -\frac{1}{2}$

$\beta = -\frac{3}{2}$

$f(x) = x^3 + 2x^2 - \frac{x}{2} - \frac{3}{2}$

(Exercise 5.5)



(a) $a(t) = -32\ ft/s^2$

$v(t) = -32t + \alpha$

$v(0) = \alpha = 128$

$v(t) = -32t + 128$

$h(t) = -16t^2 + 128t + \beta$

$h(0) = 320 = \beta$

$h(t) = -16t^2 + 128t + 320$

(b) At max height $v(t) = 0$

$t = 4\ s$

$h(4) = -16(4^2) + 128(4) + 320$

$= -256 + 512 + 320$

$h(4) = 832$

-256

$576\ ft$

(c) $h(t) = 0$ (at ground)

$16t^2 - 128t - 320 = 0$

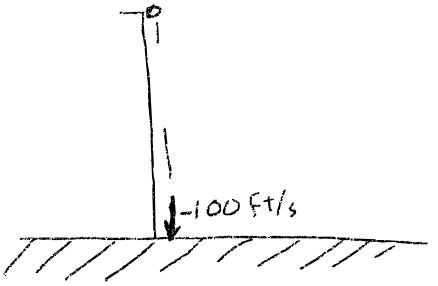
$t^2 - 8t - 20 = 0$

$t = \frac{8 \pm \sqrt{64 + 80}}{2}$

$= \frac{8 \pm 12}{2} = \underline{10\ s}$

(d) $v(10) = -320 + 128 = -192\ ft/s$

(Exercise 5.6)



$a(t) = -32 \text{ ft/s}^2$

$v(t) = -32t$

$-32t = -100$

$t = \frac{25}{8}$

$h(t) = -16t^2 + C$

$h(0) = C$

$h(t) = -16t^2 + h(0)$

$0 = -16\left(\frac{25}{8}\right)^2 + h(0)$

$h(0) = 16 \frac{25 \cdot 25}{64 \, 4} = \frac{625}{4}$

↳ 156 ft 3 inches

$25 \cdot 25$

125

50x

625

$4 \overline{)625} \; 156$

4

22

20

25

24

1

(Exercise 5.7)

$x(0) = 5$

$v(0) = 3.5$ units/s

$a(t) = \sin(2t)$

$v(t) = -\frac{1}{2}\cos(2t) + \alpha$

$v(0) = -\frac{1}{2} + \alpha = 3.5$

$\alpha = 4$

$v(t) = 4 - \frac{1}{2}\cos(2t)$

$x(t) = 4t - \frac{1}{4}\sin(2t) + \beta$

$x(0) = \beta = 5$

$x(t) = 4t - \frac{1}{4}\sin(2t) + 5$

(Exercise 5.8)

$a(t) = -20\ ft/s^2 = -6\ m/s^2$

$v(t) = (-20\ ft/s^2)\,t + \alpha$

$v(0) = 60\ mi/hr$

$= \frac{80}{3}\ m/s$

$v(t) = (-6\ m/s^2)\,t_s + \frac{80}{3}\ m/s$

$t_{stop} = \frac{80^{40}}{3 \cdot 6_3}\ s = \frac{40}{9}\ s$

$X(t) = (-3\ m/s^2)\,t^2 s^2 + \left(\frac{80}{3}\ m/s\right) t_s + \beta$

$X(0) = \beta = 0$

$X(t_{stop}) = X\left(\frac{40}{9}\right) = (-3\ m/s^2)\,\frac{40^2}{9^2}\,s^2 + \left(\frac{80}{3}\ m/s\right)\frac{40}{9}$

$X(t_{stop}) = \frac{3200}{27} - \frac{1600}{27} = \frac{1600}{27}\ m$

$\approx 59\ m$

≈ 193.6 feet

$60\ mi = 60 \times 1.6\ km$

$= 60 \times 1600\ m$

$= 96000\ m$

$60\ mi/hr = \frac{\cancel{160}^{80}\ \cancel{96000}}{\cancel{3600}_{3}}\ m/s$

$20\ ft = 20 \times 12\ in$

$= 240\ in$

$= 240 \times 2.54\ cm$

$= 2.4 \times 2.54\ m$

$\approx 6\ m$

(Exercise 5.9)

(a) Say $F(x)$ and $G(x)$ are antiderivatives of $f(x)$ and $g(x)$

So $\int f(x)\,dx = F(x) + \alpha$ and $\int g(x)\,dx = G(x) + \beta$

$$\int (f(x) + g(x))\,dx = \int (F'(x) + G'(x))\,dx$$

$$= \int (F(x) + G(x))'\,dx$$

$$= (F(x) + G(x)) + \gamma$$

$$= (F(x) + \gamma_1) + (G(x) + \gamma_2)$$

$$= \int f(x)\,dx + \int g(x)\,dx$$

$f(x) = \sin(x)$; $g(x) = \cos(x)$

$$\int [\sin(x) + \cos(x)]\,dx = \int \left[\frac{d}{dx}(-\cos(x) + \alpha) + \frac{d}{dx}(\sin(x) + \beta)\right] dx$$

$$= \int \frac{d}{dx}(-\cos x + \sin x + \alpha + \beta)\,dx$$

$$= (-\cos x + \sin x + \alpha + \beta) + \gamma$$

$$= (-\cos x + \alpha) + (\sin x + \beta) + \gamma$$

$$= \int \sin x\,dx + \int \cos x\,dx$$

(Exercise 5.9.b)

$$\int c\,f(x)\,dx = \int c\,\frac{d}{dx}\left(F(x)+\alpha\right)dx$$

$$= \int \frac{d}{dx}\left(c\,F(x)+c\alpha\right)dx$$

$$= \int \frac{d}{dx}\left(c\,F(x)\right)dx$$

$$= c\,F(x)+\beta$$

$$= c\left(F(x)+\beta'\right) \qquad (c \neq 0)$$

$$= c\int f(x)\,dx$$

Substitution

(Example 1) $\int 10x\sqrt[3]{5x^2+7}\,dx$

$u = 5x^2+7$

$du = 10x\,dx$

$= \int u^{1/3}\,du$

$= \frac{1}{1/3+1}u^{\frac{1}{3}+1} + C$

$= \frac{3}{4}\sqrt[3]{(5x^2+7)^4} + C$

$\int \tan x \sec^2 x\,dx$ $\qquad \frac{d}{dx}\sec x = \sec x \tan x$

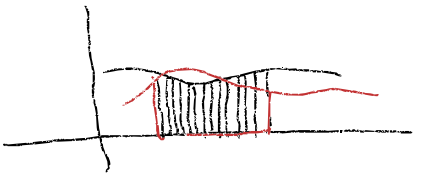
$\sec x = u$

$\sec x \tan x\,dx = du$

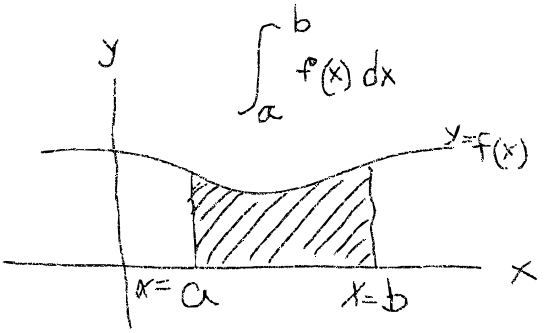
$\int u\,du = \frac{1}{2}u^2 + C$

$= \frac{1}{2}\sec^2 x + C$

(Example 2)



$$\int x e^{3x^2} dx$$



let $3x^2 = u$

$6x\,dx = du$

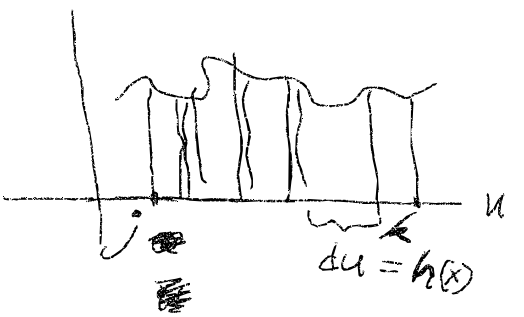
$$\frac{1}{6}\int e^u du = \frac{1}{6} e^{3x^2} + C$$

substitution
$x \mapsto u$

$g=f(u)$

(Example 3)

$$\int_1^e \frac{\sqrt{\ln x}}{x} dx$$



$\ln x = u$

$\frac{1}{x} dx = du$

$$\int_0^1 u^{1/2} du = \frac{2}{3}\left[u^{3/2}\right]_0^1 = \frac{2}{3}(1-0) = \frac{2}{3}$$

(Exercise 5.10)

(a) $\int \underbrace{\sin x}_{u} \underbrace{\cos x\, dx}_{du}$

$u = \sin x$

$du = \cos x\, dx$

$\int u \cdot du = \frac{1}{2}u^2 + C$

$= \frac{1}{2}\sin^2 x + C$

(b) $\int x\sqrt{2x^2-1}\, dx$

Let $u = 2x^2 - 1$

$du = 4x\, dx$

$\frac{1}{4}\int \sqrt{2x^2-1}\ \underbrace{4x\, dx}_{\downarrow du}$

$= \frac{1}{4}\int \sqrt{u}\, du$

$= \frac{1}{4}\cdot\frac{2}{3}u^{3/2} + C$

$= \frac{1}{6}\sqrt[2]{(2x^2-1)^3} + C$

(c) $\int \frac{24x}{(6x^2+1)^2}\, dx$

$u = 6x^2 + 1$

$du = 12x\, dx$

$2\int \frac{1}{(6x^2+1)^2}\, 12x\, dx$

$= 2\int \frac{1}{u^2}\, du$

$= 2\int u^{-2}\, du$

$= -2\frac{1}{u} + C$

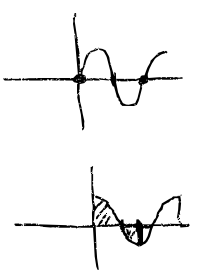
$= -\frac{2}{6x^2+1} + C$

(d) $\int_0^{\sqrt{\pi}} x\cos(x^2)\, dx$

$u = x^2$

$du = 2x\, dx$

$\frac{1}{2}\int_0^{\pi} \cos(u)\, du$



$= \frac{1}{2}\left[\sin(u)\right]_0^{\pi}$

$= \frac{1}{2}\left[\sin(\pi) - \sin(0)\right] = 0$

(Exercise 5.10)

(e) $\int_0^{\pi/4} \tan^2 x \sec^2 x \, dx$

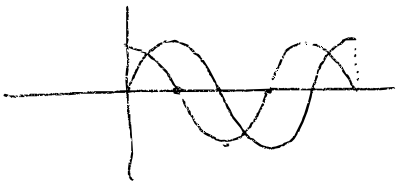
$u = \tan x$

$du = \sec^2 x \, dx$

$\int_0^1 u^2 \, du$

$= \frac{1}{3}\left[u^3\right]_0^1$

$= \frac{1}{3}$



$\tan\left(\frac{\pi}{4}\right) = \frac{\sin(\pi/4)}{\cos(\pi/4)}$

$\frac{180^\circ}{4} = 45^\circ$

(f) $\int \frac{9x^2+2}{3x^3+2x+1} \, dx$

$u = 3x^3+2x+1$

$du = (9x^2+2) \, dx$

$\int \frac{1}{u} \, du$

$= \ln u + C$

$= \ln(3x^3+2x+1) + C$

(g) $\int \frac{\ln x}{x} \, dx$

$u = \ln x$

$du = \frac{1}{x} \, dx$

$\int u \, du$

$= \frac{1}{2}u^2 + C$

$= \frac{1}{2}(\ln x)^2 + C$

(h) $\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} \, dx$

$u = \sin^{-1} x$

$du = \frac{1}{\sqrt{1-x^2}} \, dx$

$\int u \, du = \frac{1}{2}(\sin^{-1} x)^2 + C$

(Exercise 5.10)

(i) $\int \frac{\cos x}{\sqrt{1+\sin x}}\, dx$

$u = 1+\sin x$

$du = \cos x\, dx$

$\int \frac{1}{\sqrt{u}}\, du$

$= \int u^{-1/2}\, du$

$= 2u^{1/2} + c$

$= 2\sqrt{1+\sin x} + c$

(j) $\int_0^2 x e^{-x^2}\, dx$

$u = x^2$

$du = 2x\, dx$

$\frac{1}{2}\int_0^4 e^{-u}\, du$

$= -\frac{1}{2}\left[e^{-u}\right]_0^4$

$= -\frac{1}{2}(e^{-4} - 1)$

$= \frac{1}{2}(1 - e^{-4})$

(k) $\int \frac{e^x}{1+e^{2x}}\, dx$

$u = e^x$

$du = e^x dx$

$\int \frac{1}{1+u^2}\, du$

$= \tan^{-1}(u) + c$

$= \tan^{-1}(e^x) + c$

(l) $\int_{\pi/10}^{\pi/5} \cos(5x)\, dx$

$u = 5x$

$du = 5\, dx$

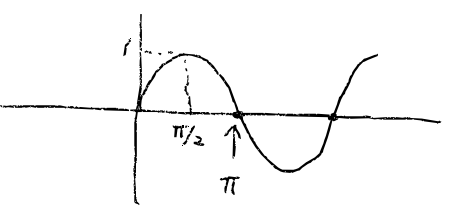
$\frac{1}{5}\int_{\frac{\pi}{2}}^{\pi} \cos(u)\, du$

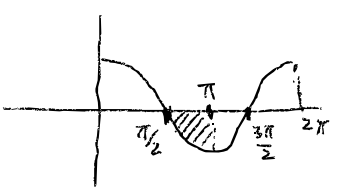
$= \frac{1}{5}\left[\sin u\right]_{\pi/2}^{\pi}$

$= \frac{1}{5}(\sin\pi - \sin\pi/2)$

$= \frac{1}{5}(0-1)$

$= -\frac{1}{5}$





Exercise (5.10)

(m) $\int_{-3}^{1} \frac{1}{\sqrt{3-2x}}\,dx$

$u = 3-2x$

$du = -2\,dx$

$-\frac{1}{2}\int_{9}^{1} \frac{1}{\sqrt{u}}\,du$

$= -\frac{1}{2}\cdot 2\left[\sqrt{u}\right]_{9}^{1}$

$= -(1-3)$

$= 2$

(n) $\int_{0}^{3} \tan^2\left(\frac{x}{3}\right)\sec^2\left(\frac{x}{3}\right)dx$

$u = \tan\left(\frac{x}{3}\right)$

$du = \frac{1}{3}\sec^2\left(\frac{x}{3}\right)dx$

$3\int_{0}^{\tan(1)} \tan^2\left(\frac{x}{3}\right)\ \frac{1}{3}\sec^2\left(\frac{x}{3}\right)dx$

$= 3\int_{0}^{\tan(1)} u^2\,du$

$= \left[u^3\right]_{0}^{\tan(1)}$

$= (\tan(1))^3$

(o) $\int \frac{\sin\sqrt{x}}{\sqrt{x}}\,dx$

$u = \sqrt{x}$

$du = \frac{1}{2\sqrt{x}}\,dx$

$2\int \frac{\sin\sqrt{x}}{2\sqrt{x}}\,dx$

$= 2\int \sin(u)\,du$

$= -2\cos(u) + C$

$= -2\cos\sqrt{x} + C$

(Exercise 5.11)

@ Bathsheba :

$$\int \sin x \cos x \, dx$$

$u = \cos x$

$du = -\sin x \, dx$

$$-\int u \, du$$

$$= -\frac{1}{2} u^2 + C$$

$$= -\frac{1}{2} \cos^2 x + C$$

$$\Downarrow$$

$$-\frac{1}{2}(1 - \sin^2 x) + C$$

$$= \frac{1}{2} + \frac{1}{2} \sin^2 x + C$$

$$= \frac{1}{2} \sin^2 x + C'$$

$$\frac{1}{2} \sin^2 x$$

$$= \frac{1}{2}(1 - \cos^2 x)$$

$$= -\frac{1}{2} \cos^2 x + \underbrace{\frac{1}{2}}_{\text{constant}}$$

If $\frac{1}{2} \sin^2 x$ is an antiderivative of $(\sin x \cos x)$ then $\frac{1}{2} \sin^2 x + C$ type all functions are valid antiderivatives.

$-\frac{1}{2} \cos^2 x$ differs from $\frac{1}{2} \sin^2 x$ by $\left(\frac{1}{2}\right)$ a constant. So it is another, valid antiderivative.

(Exercise 5·12)

$$\int \tan x \, dx$$

$$= \int \frac{\sin x}{\cos x} \, dx$$

$u = \cos x$

$du = -\sin x \, dx$

$$= -\int \frac{1}{u} \, du$$

$$= -\ln u + C$$

$$= \ln\left(\frac{1}{\cos x}\right) + C$$

$$= \ln(\sec x) + C$$

(Exercise 5·13)

$$\int \cot x \, dx$$

$$= \int \frac{\cos x}{\sin x} \, dx$$

$u = \sin x$

$du = \cos x \, dx$

$$\int \frac{1}{u} \, du$$

$$= \ln(\sin x) + C$$

(Exercise 5·14)

(a) Crosses x axis when $y = 0$

$$\sqrt{\frac{\sin\sqrt{x}}{\sqrt{x}}} = 0$$

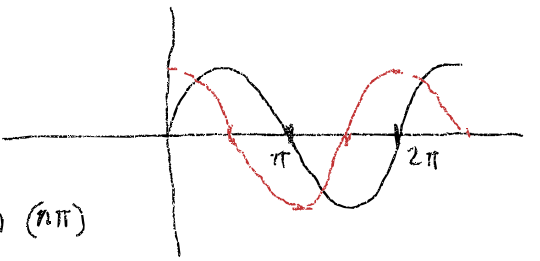
$$\frac{\sin\sqrt{x}}{\sqrt{x}} = 0$$

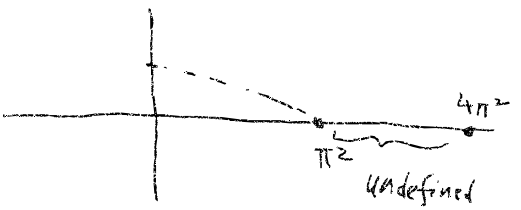
$$\sin\sqrt{x} = 0 = \sin(n\pi)$$

$$\sqrt{x} = n\pi$$

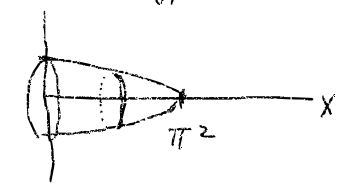
$$x = n^2\pi^2 \quad n \neq 0$$

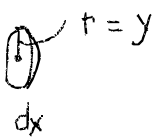
$$= \pi^2$$





(b) $y = \sqrt{\frac{\sin\sqrt{x}}{\sqrt{x}}}$





Volume of one slice $= \pi y^2 dx$

$$= \pi \frac{\sin\sqrt{x}}{\sqrt{x}} \, dx$$

Total volume $= \int_0^{\pi^2} \pi \frac{\sin\sqrt{x}}{\sqrt{x}} \, dx$

$\sqrt{x} = u$

$\frac{1}{2}\frac{1}{\sqrt{x}} dx = du$

$$= 2\pi \int_0^{\pi} \sin(u) \, du \quad = 4\pi$$

$$= -2\pi \left[\cos(u)\right]_0^{\pi}$$

$$= -2\pi \left(\cos(\pi) - 1\right)$$

$$= -2\pi(-2)$$

Exercise 5.14(c)

$$\lim_{x\to 0^+}\sqrt{\frac{\sin\sqrt{x}}{\sqrt{x}}}$$

Let's find a bit simpler one first:

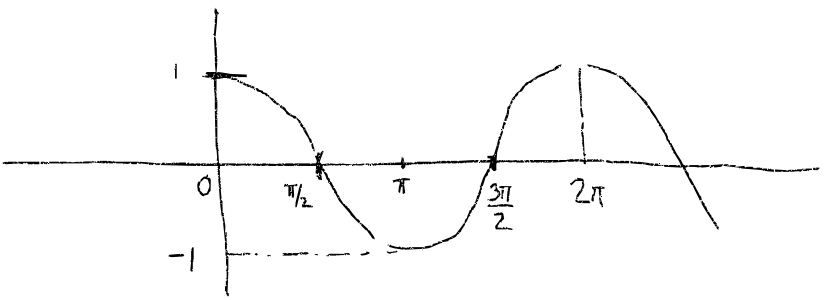
$$\lim_{x\to 0^+}\frac{\sin(\sqrt{x})}{\sqrt{x}} \overset{\text{L'Hospital}}{=} \lim_{x\to 0^+}\frac{\cos(\sqrt{x})\frac{1}{2\sqrt{x}}}{\frac{1}{2\sqrt{x}}} = \lim_{x\to 0^+}\cos(\sqrt{x})$$

$$= 1$$

So $\lim_{x\to 0^+}\sqrt{\frac{\sin\sqrt{x}}{\sqrt{x}}} = \sqrt{1} = 1$

Exercise 5.15

(a) $\int \sec x\,dx$



$$= \int \frac{1}{\cos x}\,dx$$

$$= \int \frac{\cos x}{\cos^2 x}\,dx \quad (\cos x \neq 0)$$

avoid $x = (2m+1)\pi/2$

$$= \int \frac{\cos x}{1-\sin^2 x}\,dx$$

$u = \sin x$

$du = \cos x\,dx$

$$= \int \frac{1}{1-u^2}\,du \quad (?)$$

Interlude with Elementary Functions

Derivative is easy because all elementary functions have elementary derivatives. Linearity, product and quotient rules, chain rule help. So we have an algorithm to compute derivative of any elementary function.

In contrast, there isn't a product, quotient, or chain rule for integrals. Not all elementary functions (e^x/x) have elementary antiderivatives. So, integral feels like a game, needing creativity and luck.

(Exercise 5.16)

Yes $\cosh(x) = \frac{1}{2}(e^x + e^{-x})$ is an elementary fn^c.

$$\cosh(0) = \frac{1}{2}[\exp(0) + \exp(-0)]$$

$$= \frac{1}{2}(1+1)$$

$$= 1$$

$$e^2 \cdot e^3 = e^{2+3} = e^5$$

$$\exp(2) \cdot \exp(3)$$

$$= \exp(2+3)$$

$$= \exp(5)$$

(Exercise 5.17)

$$\sinh(x) = \frac{1}{2}(e^x - e^{-x})$$

$$\sinh(0) = \frac{1}{2}(e^0 - e^0)$$

$$= 0$$

$$\frac{d}{dx}(\sinh(x)) = \frac{d}{dx}\left(\frac{1}{2}(e^x - e^{-x})\right)$$

$$\overset{\text{cons.}}{=\!=} \frac{1}{2}\frac{d}{dx}(e^x - e^{-x})$$

$$\overset{\text{lin.}}{=\!=} \frac{1}{2}\left(\frac{d}{dx}e^x - \frac{d}{dx}e^{-x}\right)$$

$$= \frac{1}{2}(e^x + e^{-x})$$

$$= \cosh(x)$$

$$\frac{d}{dx}\cosh(x)$$

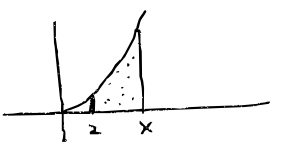
$$= \frac{d}{dx}\left(\frac{1}{2}e^x + \frac{1}{2}e^{-x}\right)$$

$$= \frac{1}{2}(e^x - e^{-x})$$

$$= \sinh(x)$$

(Exercise 5.18)

$$A_{x^2,2}(x) = \int_2^x x^2\,dx$$



$$= \frac{1}{3}[x^3]_2^x$$

$$= \frac{1}{3}(x^3 - 8)$$

Algebraic Trickery (Rational Functions)

$$\frac{x^3+5x^2+3x+5}{x^2+1}$$

$$= \frac{x(x^2+1)+5(x^2+1)+2x}{x^2+1}$$

$$= (x+5)+\frac{2x}{x^2+1}$$

$$\frac{5x+2}{(x-2)(x+2)} = \frac{A}{x-2}+\frac{B}{x+2}$$

$$5x+2 = A(x+2)+B(x-2)$$

Let $x=2$

$12 = 4A$

$A=3$

Let $x=-2$

$-8 = -4B$

$B=2$

$$\frac{5x+2}{(x-2)(x+2)} = \frac{3}{x-2}+\frac{2}{x+2}$$

Example $I = \int \frac{3x^2+5x-10}{x^2-4}\,dx$

$$\frac{3x^2+5x-10}{x^2-4} = \frac{3(x^2-4)+(5x+2)}{x^2-4}$$

$$= 3 + \frac{5x+2}{x^2-4}$$

$$I = (3x+c) + \int \frac{5x+2}{x^2-4}\,dx$$

$$\frac{5x+2}{(x+2)(x-2)} = \frac{A}{x+2} + \frac{B}{x-2}$$

$$= \frac{2}{x+2} + \frac{3}{x-2}$$

$$I = 3x + 2\ln|x+2| + 3\ln|x-2| + c$$

(Exercise 5.20)

$\ln x \equiv \log_e x = y$

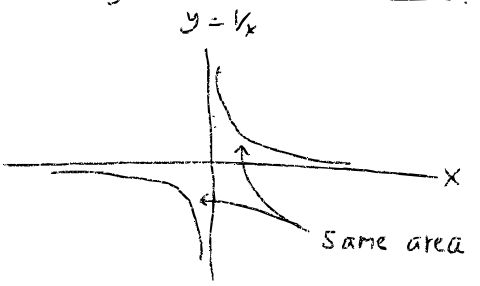
$x = e^y$

So x can never be negative or 0

$\ln x$ is defined only for $x > 0$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

But if we start with the simplest bottom-heavy rational function $\left(\frac{1}{x}\right)$ it is defined everywhere except $x = 0$.



when $x < 0$, $|x| = -x$

$$\frac{d}{dx} \ln|x| = \frac{d}{d} \ln(-x) = \frac{1}{-x}(-1) = \frac{1}{x}$$

So $\ln|x|$ is a complete antiderivative of $\frac{1}{x}$

(Exercise 5·21)

(a) $\int \frac{3}{x+5}\,dx = 3\ln|x+5| + c$

(b) $\int \frac{2}{5x-3}\,dx = \frac{2}{5}\ln|5x-3| + c$

(c) $\int_0^{\frac{e-1}{2}} \frac{3}{2x+1}\,dx = \frac{3}{2}\left[\ln|2x+1|\right]_0^{\frac{e-1}{2}}$

$= \frac{3}{2}\left(\ln|e-1+1| - \ln|1|\right)$

$= \frac{3}{2}(1-0)$

$= \frac{3}{2}$

(Exercise 5·22)

(a) $\frac{5x+7}{(x-1)(x+3)} = \frac{A}{x-1} + \frac{B}{x+3} = \frac{3}{x-1} + \frac{2}{x+3}$

$5x+7 = A(x+3) + B(x-1)$

Let $x=1$

$12 = 4A$

$A = 3$

Let $x=-3$

$-8 = -4B$

$B = 2$

(Exercise 5·22·b)

$$\frac{x}{x^2-5x+6}$$

$$= \frac{x}{(x-3)(x-2)}$$

x^2-5x+6

$= x^2-3x-2x+6$

$= x(x-3)-2(x-3)$

$= (x-3)(x-2)$

$$= \frac{-2}{x-2} + \frac{+3}{x-3}$$

(Exercise 5·22·c)

$$\frac{3x^2+8x-4}{x^3-4x} = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{x-2}$$

$$3x^2+8x-4 = A(x+2)(x-2) + Bx(x-2) + C\,x(x+2)$$

Let $x=2$

$3\cdot4+16-4 = 8C$

$24 = 8C$

$\boxed{C=3}$

Let $x=-2$

$12-16-4 = 8B$

$-8 = 8B$

$\boxed{B=-1}$

Let $x=0$

$-4 = -4A$

$\boxed{A=1}$

$$= \frac{1}{x} - \frac{1}{x+2} + \frac{3}{x-2}$$

$$= \frac{2}{x(x+2)} + \frac{3}{x-2}$$

$$= \frac{2x-4+3x(x+2)}{x(x+2)(x-2)}$$

$$= \frac{2x+4+3x^2+6x}{x^3-4x}$$

(Exercise 5·23·a)

$$\frac{x+4}{x-1} = \frac{x-1+5}{x-1} = 1 + \frac{5}{x-1}$$

(Exercise 5·23·b)

$$\frac{6x^2-2}{3x^2-x+1} = \frac{3x^2-x+1+(3x^2+x-3)}{3x^2-x+1}$$

$$= 1 + \frac{(3x^2-x+1)+(2x-4)}{3x^2-x+1}$$

$$= 2 + \frac{2x-4}{3x^2-x+1}$$

(Exercise 5·23·c)

$$\frac{8x^4+2x^2+1}{2x^2+3} = \frac{4x^2(2x^2+3)-10x^2+1}{2x^2+3}$$

$$= \frac{2x^2(4x^2+1)+1}{2x^2+3}$$

$$= \frac{2x^2(2x^2+3)+2x^2(2x^2+3)-10x^2+1}{2x^2+3}$$

$$= 4x^2 - \frac{10x^2-1}{2x^2+3}$$

$$= 4x^2 - \frac{5(2x^2+3)-16}{2x^2+3}$$

$$= (4x^2-5) + \frac{16}{2x^2+3}$$

$$\frac{(4x^2-5)(2x^2+3)+16}{2x^2+3}$$

$$= \frac{8x^4+12x^2-10x^2-15+16}{2x^2+3}$$

$$= \frac{8x^4+2x^2+1}{2x^2+3}$$

(Exercise 5.24)

false premise

$$\frac{x^2+1}{(x-1)x} = \frac{A}{x-1} + \frac{B}{x} = \frac{2}{x-1} - \frac{1}{x}$$

$$x^2+1 = Ax + B(x-1)$$

Let $x=1$

$$2 = A$$

Let $x=0$

$$1 = -B$$

$$B = -1$$

(Exercise 5.25.a)

$$\frac{5x+7}{(x-1)(x+3)} = \frac{A}{x-1} + \frac{B}{x+3} = \frac{3}{x-1} + \frac{2}{x+3}$$

$$5x+7 = A(x+3) + B(x-1)$$

Let $x=1$

$$12 = 4A$$

$$A = 3$$

Let $x=-3$

$$-8 = -4B$$

$$B = 2$$

$$\int \frac{5x+7}{(x-1)(x+3)}\,dx = 3\ln|x-1| + 2\ln|x+3| + C$$

(Exercise 5.25.b)

$$\frac{10-2x}{x(x+5)} = \frac{A}{x} + \frac{B}{x+5} = \frac{2}{x} - \frac{4}{x+5}$$

$$10-2x = A(x+5) + Bx$$

Let $x=0$

$$10 = 5A$$

$$A = 2$$

Let $x=-5$

$$10+10 = -5B$$

$$B = -4$$

$$\int \frac{10-2x}{x^2+5x}\,dx = 2\ln|x| - 4\ln|x+5| + c$$

(Exercise 5.25.c)

$$\frac{x^4}{x^2-4} = \frac{x^2(x^2-4)+4x^2}{x^2-4} = x^2 + \frac{4x^2}{x^2-4}$$

$$\int \frac{x^4}{x^2-4}\,dx$$

$$= \frac{1}{3}x^3 + 4x + 4\ln|x-2| - 4\ln|x+2| + c$$

$$= \frac{1}{3}x^3 + 4x + 4\ln\frac{|x-2|}{|x+2|} + c$$

$$= x^2 + \frac{4(x^2-4)+16}{x^2-4}$$

$$= (x^2+4) + \frac{16}{x^2-4}$$

$$\frac{16}{(x+2)(x-2)} = \frac{4}{x-2} - \frac{4}{x+2}$$

(Exercise 5.26)

$$\frac{1}{P(1-mP)} = \frac{(1-mP)+mP}{P(1-mP)}$$

$$= \frac{1}{P} + \frac{m}{1-mP}$$

$$\int \frac{1}{P(1-mP)}\,dP = \ln|P| - \ln|1-mP| + C$$

$$= \ln\frac{|P|}{|1-mP|} + C$$

$$= \ln\left|\frac{P}{1+mP}\right| + C$$

(Exercise 5.27)

(a) $\frac{d}{dx}\left(\ln|\sec x + \tan x| + c\right)$

$= \frac{1}{\sec x + \tan x}\left(\sec x \tan x + \sec^2 x\right)$

$= \sec x$

(b) $\int \sec x \, dx$

$= \int \frac{1}{\cos x} \, dx$

$= \int \frac{\cos x}{\cos^2 x} \, dx$

$= \int \frac{\cos x}{1 - \sin^2 x} \, dx$

let $\sin x = u$

$\cos x \, dx = du$

$= \int \frac{du}{1 - u^2}$

$= \int \frac{1}{2}\left(\frac{1}{1-u} + \frac{1}{1+u}\right) du$

$= \frac{1}{2}\left(\ln|1-u|(-1) + \ln|1+u|\right) + c$

$= \frac{1}{2} \ln\left|\frac{1+u}{1-u}\right| + c$

$\frac{d}{dx} \ln|u(x)| = \frac{d}{dx} \ln(u(x))$

$= \frac{u'(x)}{u(x)}$

When $u(x) > 0$, $|u(x)| = u(x)$

When $u(x) < 0$, $|u(x)| = -u(x)$

$\frac{d}{dx} \ln|u(x)| = \frac{d}{dx} \ln(-u(x))$

$= \frac{1}{-u(x)} \cdot -u'(x)$

$= \frac{u'(x)}{u(x)}$

$\frac{1}{1-u^2} = \frac{A}{1+u} + \frac{B}{1-u}$

$1 = A(1-u) + B(1+u)$

$u = 1$

$1 = 2B$

$B = \frac{1}{2}$

$u = -1$

$1 = +2A$

$A = +\frac{1}{2}$

$\frac{1}{1-u^2} = \frac{1}{2}\left(\frac{1}{1-u} + \frac{1}{1+u}\right)$

5.27

(b) continued...

$$\int \sec x \, dx = \frac{1}{2} \ln \left| \frac{1+\sin x}{1-\sin x} \right| + c$$

$$= \frac{1}{2} \ln \frac{(1+\sin x)^2}{\cos^2 x} + c$$

$$= \ln \left| \frac{1+\sin x}{\cos x} \right| + c$$

$$= \ln |\sec x + \tan x| + c$$

(Exercise 5.28)

$$\frac{d}{dx}\left(F(x) + c\right) = \operatorname{cosec}(x)$$

$$\int \csc(x) \, dx$$

$$= \int \frac{1}{\sin x} \, dx$$

$$= \int \frac{\sin x}{1-\cos^2 x} \, dx$$

Let $\cos x = u$

$-\sin x \, dx = du$

$$= -\int \frac{du}{1-u^2}$$

$$= -\frac{1}{2} \int \left(\frac{1}{1+u} + \frac{1}{1-u} \right) du$$

$$= -\frac{1}{2} \left(\ln|1+u| - \ln|1-u| \right) + c$$

$$= \frac{1}{2} \ln \left| \frac{1-u}{1+u} \right| + c$$

$$= \frac{1}{2} \ln \left| \frac{1-\cos x}{1+\cos x} \right| + c$$

$$= \frac{1}{2} \ln \left(\frac{1-\cos x}{\sin x} \right)^2 + c$$

$$= \ln \left| \frac{1-\cos x}{\sin x} \right| + c$$

$$= \ln |\csc x - \cot x| + c$$

Trigonometric Trickery (Part 1)

$\cos^2 x + \sin^2 x = 1$

$\cos^2 x - \sin^2 x = \cos(2x)$

$1 + \tan^2 x = \sec^2 x$

$2 \sin x \cos x = \sin(2x)$

$2\cos^2 x = 1 + \cos(2x)$

$\cos^2 x = \frac{1 + \cos(2x)}{2}$

$1 - \sin^2 x = \frac{1 + \cos(2x)}{2}$

$\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos(2x)$

$= \frac{1}{2}\left(1 - \cos(2x)\right)$

Example 1 $\int \cos^2(3x)\, dx$

$= \frac{1}{2}\int \left(1 + \cos(6x)\right) dx$

$= \frac{1}{2}x + \frac{1}{12}\sin(6x) + C$

Example 2 $\int \sin^4 x \cos^5 x \, dx$

$= \int \sin^4 x \cos^4 x \cos x \, dx$

$= \int \sin^4 x (1 - \sin^2 x)^2 \cos x \, dx$

$= \int u^4 (1 - u^2)^2 \, du$

$= \int u^4 (1 - 2u^2 + u^4) \, du$

$= \int (u^4 - 2u^6 + u^8) \, du$

$= \frac{1}{5}u^5 - \frac{2}{7}u^7 + \frac{1}{9}u^9 + C$

$= \frac{1}{5}\sin^5 x - \frac{2}{7}\sin^7 x + \frac{1}{9}\sin^9 x + C$

(Exercise 5·29)

$\cos^2\theta + \sin^2\theta = 1$

$$\frac{\cos^2\theta + \sin^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta} \quad [\cos\theta \neq 0]$$

$$1 + \tan^2\theta = \sec^2\theta$$

(Exercise 5·30)

Half angle identities:

$$\cos^2\theta = \frac{1+\cos(2\theta)}{2} \qquad \sin^2\theta = \frac{1-\cos(2\theta)}{2}$$

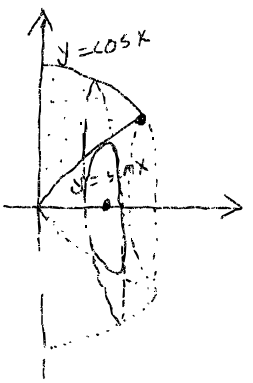
$$\begin{aligned}\cos(2\theta) &= \cos^2\theta - \sin^2\theta \\ &= \cos^2\theta - (1-\cos^2\theta) \\ &= \cos^2\theta - 1 + \cos^2\theta\end{aligned}$$

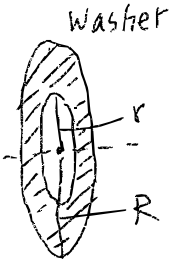
$$1 + \cos(2\theta) = 2\cos^2\theta$$

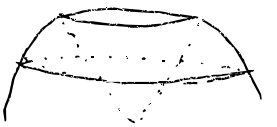
$$\cos^2\theta = \frac{1+\cos(2\theta)}{2}$$

(Exercise 5·31)

(a)









$\cos x = \sin x$

$\cos^2 x = \sin^2 x$

$\cos 2x = 0 = \cos \pi/2$

$x = \frac{\pi}{4}$

(b) $r(x) = \sin x$

$R(x) = \cos x$

area of washer $= \pi R^2(x) - \pi r^2(x)$

$= \pi(\cos^2 x - \sin^2 x)$

$= \pi \cos(2x)$

volume of washer $= \pi \cos(2x)\, dx$

(c) $V = \int_0^{\pi/4} \pi \cos(2x)\, dx$

(d) $= \frac{\pi}{2} [\sin 2x]_0^{\pi/4}$

$= \frac{\pi}{2} (\sin \pi/2 - \sin 0)$

$= \frac{\pi}{2}$ unit^3

(Exercise 5.32)

(a) $\int \cos^2 x \, dx$

$= \frac{1}{2}\int \left(1 + \cos(2x)\right) dx$

$= \frac{1}{2}\left(x + \frac{1}{2}\sin(2x)\right) + C$

$= \frac{1}{2}x + \frac{1}{4}\sin(2x) + C$

(b) $\int_{\pi/8}^{\pi/4} 4\sin^2 x \, dx$

$= \frac{4}{2}\int_{\pi/8}^{\pi/4} \left[1 - \cos(2x)\right] dx$

$= 2\left[x - \frac{1}{2}\sin(2x)\right]_{\frac{\pi}{8}}^{\pi/4}$

$= 2\left[\frac{\pi}{4} - \frac{1}{2}\sin\left(\frac{\pi}{2}\right) - \frac{\pi}{8} + \frac{1}{2}\sin\left(\frac{\pi}{4}\right)\right]$

$= 2\left(\frac{\pi}{8} - \frac{1}{2} + \frac{1}{2\sqrt{2}}\right)$

$= \frac{\pi}{4} - 1 + \frac{1}{\sqrt{2}}$

(c) $\int \cos^2(2\theta) \, d\theta$

$= \frac{1}{2}\int \left[1 + \cos(4\theta)\right] d\theta$

$= \frac{1}{2}\left[\theta + \frac{1}{4}\sin(4\theta)\right] + C$

$= \frac{\theta}{2} + \frac{1}{8}\sin(4\theta) + C$

(d) $\int_{\pi/12}^{\pi/4} \sin^2(4x) \, dx$

$= \frac{1}{2}\int_{\pi/12}^{\pi/4} \left[1 - \cos(8x)\right] dx$

$= \frac{1}{2}\left[x - \frac{1}{8}\sin(8x)\right]_{\frac{\pi}{12}}^{\frac{\pi}{4}}$

$= \frac{1}{2}\left[\frac{\pi}{4} - \frac{1}{8}\sin(2\pi) - \frac{\pi}{12} + \frac{1}{8}\sin\left(\frac{2\pi}{3}\right)\right] + C$

$= \frac{\pi}{8} - \frac{1}{16}\sin(2\pi) - \frac{\pi}{24} + \frac{1}{16}\sin\left(\frac{2\pi}{3}\right) + C$

$= \frac{\pi}{12} + \frac{1}{16}\left[\sin\left(\frac{2\pi}{3}\right) - \sin(2\pi)\right] + C$

$= \frac{\pi}{12} + \frac{\sqrt{3}}{32} + C$

Exercise 5.32 cont.

(e) $\int (\cos^2 x - \sin^2 x)\,dx$

$= \int \cos(2x)\,dx$

$= \frac{1}{2}\sin(2x) + C$

(f) $\int_{\pi/4}^{\pi/3} (1 + \tan^2 x)\,dx$

$= \int_{\pi/4}^{\pi/3} \sec^2 x\,dx$

$= [\tan x]_{\pi/4}^{\pi/3}$

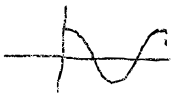
$= \tan(\frac{\pi}{3}) - \tan(\frac{\pi}{4})$

$= \sqrt{3} - 1$

(g) $\int_0^{\pi} \sin x \cos x\,dx$

$= \frac{1}{2}\int_0^{\pi} \sin(2x)\,dx$

$= -\frac{1}{4}[\cos(2x)]_0^{\pi}$



$= -\frac{1}{4}[\cos(2\pi) - \cos(0)] = 0$

(h) $\int \cos^2 x \sin^3 x\,dx$

$= \int \cos^2 x \sin^2 x \sin x\,dx$

$= \int \cos^2 x (1 - \cos^2 x) \sin x\,dx$

Let $\cos x = u$

$-\sin x\,dx = du$

$= -\int u^2(1-u^2)\,du$

$= \int (-u^2 + u^4)\,du$

$= -\frac{1}{3}u^3 + \frac{1}{5}u^5 + C$

$= \frac{1}{5}\cos^5 x - \frac{1}{3}\cos^3 x + C$

(i) $\int \sin^3 x \cos^3 x\,dx$

$= \int \sin^3 x \cos^2 x \cos x\,dx$

$= \int u^3(1-u^2)\,du$

$= \frac{1}{4}u^4 - \frac{1}{6}u^6 + C$

$= \frac{1}{4}\sin^4 x - \frac{1}{6}\sin^6 x + C$

(i) $\int \sin^3 x \cos^6 x\,dx$

$= \int \cos^6 x (1 - \cos^2 x) \sin x\,dx$

$= -\int u^6(1-u^2)\,du$

$= \int (-u^6 + u^8)\,du$

$= -\frac{1}{7}\cos^7 x + \frac{1}{9}\cos^9 x + C$

(k) $\int \sin^2 x \cos^2 x\,dx$

$= \frac{1}{4}\int (2\sin x \cos x)^2\,dx$

$= \frac{1}{4}\int \sin^2(2x)\,dx$

$= \frac{1}{4}\cdot\frac{1}{2}\int (1 - \cos(4x))\,dx$

$= \frac{1}{8}[x - \frac{1}{4}\sin(4x)] + C$

(l) $\int \tan^3 x \sec^6 x\,dx$

$= \int \tan^3 x \sec^4 x \sec^2 x\,dx$

$= \int \tan^3 x (1 + \tan^2 x)^2 \sec^2 x\,dx$

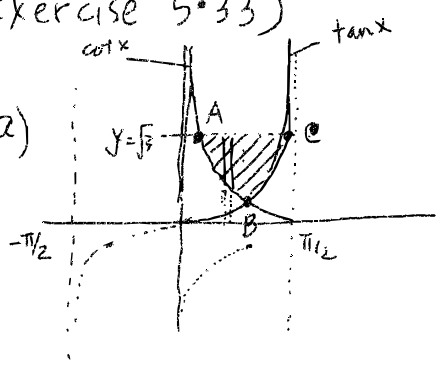
$= \int u^3(1+u^2)^2\,du$

$= \int u^3(1 + 2u^2 + u^4)\,du$

$= \frac{1}{4}\tan^4 x + \frac{1}{3}\tan^6 x + \frac{1}{8}\tan^8 x + C$

(Exercise 5.33)

(a)



(b) $\tan x = \cot x$

$\frac{\sin x}{\cos x} = \frac{\cos x}{\sin x}$

$\cos^2 x - \sin^2 x = 0$

$\cos(2x) = 0 = \cos(\pi/2)$

$x = \pi/4$

$B(\frac{\pi}{4}, 1)$

$\tan x = \sqrt{3}$

$x = \frac{\pi}{3}$

$C(\frac{\pi}{3}, \sqrt{3})$

$\cot x = \sqrt{3}$

$\tan x = \frac{1}{\sqrt{3}}$

$x = \frac{\pi}{6}$

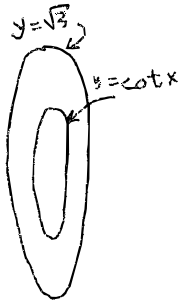
$A(\frac{\pi}{6}, \sqrt{3})$

$\tan x = \frac{\sin x}{\cos x}$

(c)

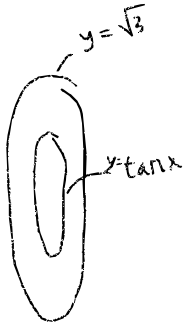


(d)



$\frac{\pi}{6} \leq x \leq \frac{\pi}{4}$

Volume $= \pi(3 - \cot^2 x)\,dx$



$\frac{\pi}{4} \leq x \leq \frac{\pi}{3}$

Volume $= \pi(3 - \tan^2 x)\,dx$

(e) $V = 2\pi \int_{\pi/4}^{\pi/3} (3 - \tan^2 x)\,dx$

(f) $V = 2\pi\left[3x\right]_{\pi/4}^{\pi/3} - 2\pi \int_{\pi/4}^{\pi/3} \tan^2 x\,dx$

$= 2\pi\left(\pi - \frac{3\pi}{4}\right) - 2\pi \int_{\pi/4}^{\pi/3} (\sec^2 x - 1)\,dx$

$= 2\pi\left(\frac{\pi}{4}\right) - 2\pi \int_{\pi/4}^{\pi/3} \sec^2 x\,dx + 2\pi\left[x\right]_{\pi/4}^{\pi/3}$

$= \frac{\pi^2}{2} - 2\pi\left[\tan x\right]_{\pi/4}^{\pi/3} + 2\pi\left(\frac{\pi}{3} - \frac{\pi}{4}\right)$

$= \frac{\pi^2}{2} - 2\pi\left(\sqrt{3} - 1\right) + 2\pi \cdot \frac{\pi}{12}$

$= \pi^2/2 - 2(\sqrt{3}-1)\pi + \frac{\pi^2}{6} = \frac{2}{3}\pi^2 - 2(\sqrt{3}-1)\pi$

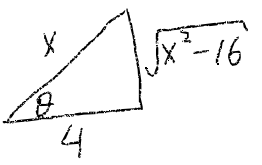
Example $\int \frac{\sqrt{x^2-16}}{x}\,dx$

$\sec^2\theta - 1 = \tan^2\theta$

$x = 4\sec\theta \Rightarrow \cos\theta = \frac{4}{x}$

$\frac{dx}{d\theta} = 4\sec\theta\tan\theta$

$dx = 4\sec\theta\tan\theta\,d\theta$



$\tan\theta = \frac{\sqrt{x^2-16}}{4}$

$$\int \frac{\sqrt{16\sec^2\theta - 16}\ 4\sec\theta\tan\theta\,d\theta}{4\sec\theta}$$

$= 4\int \tan^2\theta\,d\theta$

$= 4\int(\sec^2\theta - 1)\,d\theta$

$= 4\int\sec^2\theta\,d\theta - 4\theta + C$

$= 4\tan\theta - 4\theta + C$ ✓

$= \sqrt{x^2-16} - 4\tan^{-1}\frac{\sqrt{x^2-16}}{4} + C$

(Exercise 5.34)

(a) $x = 3\sin\theta$

$9 - x^2$

$= 9 - 9\sin^2\theta$

$= 9(1-\sin^2\theta)$

$= (3\cos\theta)^2$

(b) $x = \sqrt{2}\sin\theta$

$(\sqrt{2}\cos\theta)^2$

(c) $x^2 + 16$

$x = 4\tan\theta$

$16\tan^2\theta + 16$

$16\sec^2\theta = (4\sec\theta)^2$

(d) $x^2 + 15$

$x = \sqrt{15}\tan\theta$

$15(\tan^2\theta + 1)$

$= (\sqrt{15}\sec\theta)^2$

(e) $x^2 - 3$

$x = \sqrt{3}\sec\theta$

$3(\sec^2\theta - 1) = (\sqrt{3}\tan\theta)^2$

(f) $9x^2 + 25$

$x = \frac{5}{3}\tan\theta$

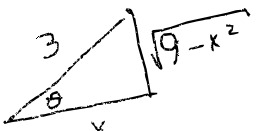
$25(\tan^2\theta + 1)$

$= 25\sec^2\theta$

$= (5\sec\theta)^2$

(Exercise 5.35)

(a) $\int \frac{\sqrt{9-x^2}}{x^2}\,dx$



Let $x = 3\cos\theta$

$\frac{dx}{d\theta} = -3\sin\theta$

$dx = -3\sin\theta\,d\theta$

$\tan\theta = \frac{\sqrt{9-x^2}}{x}$

$\sin\theta = \frac{\sqrt{9-x^2}}{3}$

$\sqrt{9-x^2} = \sqrt{9-9\cos^2\theta}$

$= 3\sin\theta$

$$\int \frac{3\sin\theta}{9\cos^2\theta}(-3\sin\theta\,d\theta)$$

$$= -\int \tan^2\theta\,d\theta$$

$$= \int (1-\sec^2\theta)\,d\theta$$

$$= \theta - \tan\theta + C$$

$$= \cos^{-1}(x/3) - \frac{\sqrt{9-x^2}}{x} + C$$

(Exercise 5.35.b)

$$\int \frac{x^2}{\sqrt{9-x^3}}\,dx$$

$\sqrt{9-x^3} = u$

$\frac{1}{2\sqrt{9-x^3}} \cdot -3x^2 = \frac{du}{dx}$

$-\frac{3}{2}\frac{x^2}{\sqrt{9-x^3}}\,dx = du$

$$\int \left(-\frac{2}{3}\right) du$$

$= -\frac{2}{3}\sqrt{9-x^3} + C$

$= \int \frac{\cos\theta}{1-\sin^2\theta}\,d\theta$

$u = \sin\theta$

$du = \cos\theta\,d\theta$

$= \int \frac{du}{1-u^2}$

$= \frac{1}{2}\left(\ln|1+u| - \ln|u-1|\right) + C$

$= \frac{1}{2}\ln\left|\frac{1+u}{1-u}\right| + C$

$= \frac{1}{2}\ln\left|\frac{1+\sin\theta}{1-\sin\theta}\right| + C$

$= \frac{1}{2}\ln\left|\frac{(1+\sin\theta)}{\cos\theta}\right|^2 + C$

$= \ln|\sec\theta + \tan\theta| + C$

$= \ln\left|\frac{\sqrt{x^2+4}}{2} + \frac{x}{2}\right| + C$

$= \underline{\ln\left|\sqrt{x^2+4} + x\right| + C}$

$\frac{1}{1-u^2} = \frac{A}{1+u} + \frac{B}{1-u}$

$1 = A(1-u) + B(1+u)$

$u = 1$

$1 = 2B \Rightarrow B = 1/2$

$u = -1$

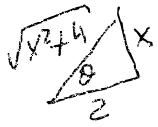
$1 = +2A \Rightarrow A = +\frac{1}{2}$

$\frac{1}{1-u^2} = \frac{1}{2}\left(\frac{1}{1+u} + \frac{1}{1-u}\right)$

$= \frac{1}{2}\left(\frac{1}{1+u} - \frac{1}{u-1}\right)$

(Exercise 5.35.c)

$$\int \frac{1}{\sqrt{4+x^2}}\,dx$$



$x = 2\tan\theta$

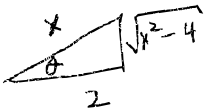
$dx = 2\sec^2\theta\,d\theta$

$\int \frac{2\sec^2\theta}{2\sec\theta}\,d\theta$

$= \int \sec\theta\,d\theta$

(Exercise 5.35.d)

$$\int \frac{x}{\sqrt{x^2-4}}\,dx$$



$x = 2\sec\theta$

$dx = 2\sec\theta\tan\theta\,d\theta$

$$\int \frac{2\sec\theta}{2\tan\theta}\,2\sec\theta\tan\theta\,d\theta$$

$$= 2\int \sec^2\theta\,d\theta$$

$$= 2\tan\theta + C$$

$$= 2\,\frac{\sqrt{x^2-4}}{2} + C$$

$$= \sqrt{x^2-4} + C$$

(Exercise 5.35.e)

$$\int \frac{1}{\sqrt{16-x^2}}\,dx$$

$x = 4\sin\theta$

$dx = 4\cos\theta\,d\theta$

$$\int \frac{4\cos\theta\,d\theta}{4\cos\theta}$$

$$= \theta + C$$

$$= \sin^{-1}(x/4) + C$$

(Exercise 5.36)

$$\ln\left|\frac{x}{3}\right| + C$$

$$= \ln\frac{|x|}{|3|} + C$$

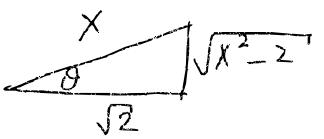
$$= \ln|x| + \ln|3| + C$$

$$= \ln|x| + C'$$

(Exercise 5.37)

$$\int \frac{1}{\sqrt{x^2-2}}\,dx$$

$x = \sqrt{2}\sec\theta$



$dx = \sqrt{2}\sec\theta\tan\theta\,d\theta$

$$\sqrt{2}\int \frac{\sec\theta\,\cancel{\tan\theta}}{\sqrt{2}\,\cancel{\tan\theta}}\,d\theta$$

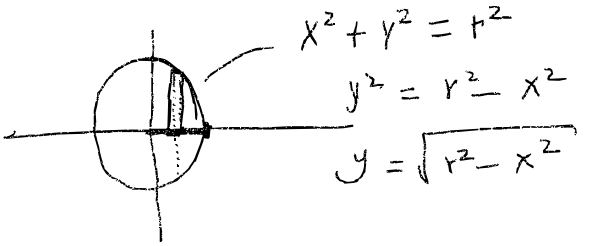
$$= \int \sec\theta\,d\theta$$

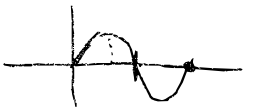
$$= \ln\left|\sec\theta + \tan\theta\right| + C$$

$$= \ln\left|\frac{x}{\sqrt{2}} + \frac{\sqrt{x^2-2}}{\sqrt{2}}\right| + C$$

$$= \ln\left|x + \sqrt{x^2-2}\right| + C$$

(Exercise 5.38)



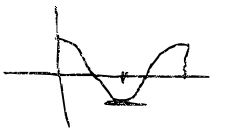


Area of first quadrant $= \int_0^r \sqrt{r^2 - x^2}\, dx$

Let $x = r\sin\theta$

$dx = r\cos\theta\, d\theta$

$= \int_0^{\pi/2} r^2\cos^2\theta\, d\theta$



$\cos 2\theta = \cos^2\theta - \sin^2\theta$

$= 2\cos^2\theta - 1$

$= \frac{r^2}{2} \int_0^{\pi/2} \left[1 + \cos(2\theta)\right] d\theta$

$= \frac{r^2}{2} \left[\theta + \frac{1}{2}\sin(2\theta)\right]_0^{\pi/2}$

$= \frac{r^2}{2}\left(\frac{\pi}{2} + \frac{1}{2}\sin(\pi)^{0} - 0\right)$

$= \frac{\pi}{4} r^2$

So total area $= 4\left(\frac{\pi}{4} r^2\right)$

$= \pi r^2$

(Exercise 5.39)

$$\int \frac{1}{16+x^2}\,dx$$

$x = 4\tan\theta$

$dx = 4\sec^2\theta\, d\theta$

$$\int \frac{1}{16+16\tan^2\theta}\,4\sec^2\theta\, d\theta$$

$$= \frac{1}{4}\int 1\, d\theta$$

$$= \frac{1}{4}\theta + C$$

$$= \frac{1}{4}\tan^{-1}\left(\frac{x}{4}\right) + C$$

Integration by parts

Example. $\int x \cos x \, dx$ ($x \to 1$, $\cos x \to \sin x$)

$= x \sin x - \int 1 \cdot \sin x \, dx$

$= x \sin x + \cos x + C$

Example $\int x^2 e^x \, dx$ ($x^2 \to 2x$, $e^x \to e^x$)

$= x^2 e^x - 2\int x e^x \, dx$ ($x \to 1$, $e^x \to e^x$)

$= x^2 e^x - 2\left[x e^x - \int 1 \cdot e^x \, dx\right]$

$= x^2 e^x - 2x e^x + 2e^x + C$

$\int u \frac{dv}{dx} \, dx = uv - \int \frac{du}{dx} \cdot v \, dx$ ($v \to \frac{dv}{dx}$, $u \to du/dx$)

$\frac{d}{dx}(uv) = \frac{du}{dx} v + u \frac{dv}{dx}$

$uv = \int \frac{du}{dx} \cdot v \, dx + \int u \cdot \frac{dv}{dx} \, dx$

(Problem 5.40)

$$\frac{d}{dx}(uv) = \frac{du}{dx}\cdot v + u\cdot\frac{dv}{dx}$$

$$\int \frac{d}{dx}(uv)\,dx = \int \frac{du}{dx}\cdot v\,dx + \int u\,\frac{dv}{dx}\,dx$$

$$\int u\left(\frac{dv}{dx}\right)dx = uv - \int\left(\frac{du}{dx}\right)\cdot v\,dx$$

(a) $\int_{u} \underset{\downarrow}{x}\,\underset{\downarrow}{e^x}_{\frac{dv}{dx}}\,dx$ (e^x → v, x → u, 1 → $\frac{du}{dx}$)

$= xe^x - \int 1\cdot e^x\,dx$

$= xe^x - e^x + C$

$= (x-1)e^x + C$

(b) $\int \underset{\frac{dv}{dx}}{x^9}\,\underbrace{\ln x}_{u}\,dx$ ($\frac{1}{10}x^{10}$ → v, x^{10} → $\frac{dv}{dx}$)

$= \ln x\left(\frac{1}{10}x^{10}\right) - \int \frac{1}{x}\cdot\frac{1}{10}\cdot x^{10}\,dx$

$= \frac{1}{10}x^{10}\ln x - \frac{1}{10}\int x^9\,dx$

$= \frac{1}{10}x^{10}\ln x - \frac{1}{100}x^{10} + C$

$= \frac{x^{10}}{100}\left(10\ln x - 1\right) + C$ ✓

Check:

$\frac{1}{10}x^9(10\ln x - 1) + \frac{10}{x}\cdot\frac{x^{10}}{100}$

$= x^9\ln x - \frac{x^9}{10} + \frac{x^9}{10}$

$= x^9\ln x$

(Problem 5.40.c)

$$\int \underset{u}{x^2} \underset{dv/du}{\sin x}\, dx$$

$$= -x^2 \cos x + \int \underset{u}{2x} \underset{dv/dx}{\cos x}\, dx$$

$$= -x^2 \cos x + 2\left(x \sin x - \int 1 \cdot \sin x\, dx \right)$$

$$= -x^2 \cos x + 2x \sin x + 2\cos x + C$$

(Problem 5.40.d)

$$\int \underset{u}{x} \underset{dv/dx}{\sin(3x+2)}\, dx$$

$$= x\left(-\tfrac{1}{3}\cos(3x+2)\right) - \int 1 \cdot \left(-\tfrac{1}{3}\cos(3x+2)\right) dx$$

$$= -\frac{x}{3}\cos(3x+2) + \frac{1}{3}\int \cos(3x+2)\, dx$$

$$= -\frac{x}{3}\cos(3x+2) + \frac{1}{9}\sin(3x+2) + C$$

(Problem 5.40.e)

$$\int \frac{\ln(\ln x)}{x}\, dx$$

Note, $\ln(\ln x)$ is defined when $\ln x > 0$ or $x > 1$ so for this problem $\ln|x| = \ln(x)$

$$= \int \underbrace{\ln(\ln x)}_{u} \cdot \underbrace{\frac{1}{x}}_{dv/dx}\, dx$$

$$= \ln(\ln x) \cdot \ln x - \int \frac{1}{\ln x}\frac{1}{x} \ln x\, dx$$

$$= \ln x \cdot \ln(\ln x) - \ln(x) + C$$

$$= \ln(x)\left[\ln(\ln x) - 1\right] + C$$

(Problem 5.41.a)

$$\int \ln x \, dx \; ; \; \underline{x>0}$$

$$= \int \underbrace{1}_{\frac{dy}{dx}} \cdot \underbrace{\ln x}_{u} \, dx$$

$$= x\ln x - \int \frac{1}{x} \cdot x \, dx$$

$$= x\ln x - x + C$$

$$= x(\ln x - 1) + C$$

(Problem 5.41.b)

$$\int 1 \cdot \underbrace{\tan^{-1} x}_{u} \, dx$$

$$= x\tan^{-1}x - \underbrace{\int \frac{1}{1+x^2} \, x \, dx}_{\text{let } 1+x^2 = u, \; 2x\,dx = du}$$

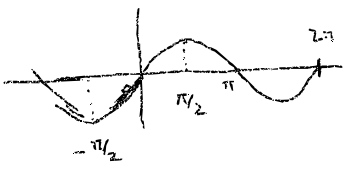
$$= x\tan^{-1}(x) - \frac{1}{2}\int \frac{1}{u} \, du$$

if $x>0$, $u>1$

$$= x\tan^{-1}(x) - \frac{1}{2}\ln(u) + C$$

$$= x\tan^{-1}(x) - \ln\sqrt{1+x^2} + C$$

(Problem 5.41.c)



$$\int 1 \cdot \underbrace{\sin^{-1}(x)}_{u}\, dx$$

$$= x\sin^{-1}(x) - \underbrace{\int \frac{1}{\sqrt{1-x^2}}\, x\, dx}$$

$1 - x^2 = u \qquad u \in [-1, 1$

$-2x\, dx = du$

$$= x\sin^{-1}(x) + \frac{1}{2}\int \frac{-2x\, dx}{\sqrt{1-x^2}}$$

$$= x\sin^{-1}(x) + \frac{1}{2}\int \frac{1}{\sqrt{u}}\, du$$

$$= x\sin^{-1}(x) + \frac{1}{2}\,\frac{1}{-\frac{1}{2}+1}\, u^{-\frac{1}{2}+1} + C$$

$$= x\sin^{-1}(x) + \sqrt{1-x^2} + C$$

$1 - x^2 \geq 0$

$x^2 \leq 1$

$-1 \leq x \leq 1$

(Problem 5.41.d)

$$\int 1 \cdot \underbrace{\cos^{-1}(x)}_{u}\, dx$$

$$= x\cos^{-1}(x) - \int -\frac{1}{\sqrt{1-x^2}}\, x\, dx$$

$$= x\cos^{-1}(x) - \frac{1}{2}\int \frac{-2x\, dx}{\sqrt{1-x^2}}$$

$$= x\cos^{-1}(x) - u^{1/2} + C$$

$$= x\cos^{-1}(x) - \sqrt{1-x^2} + C$$

(Problem 5.42)

$$I = \int \underbrace{e^x}_{dv/dx} \underbrace{\cos x}_{u}\, dx$$

$$= e^x \cos x + \int \overbrace{e^x}^{dv/dx} \overbrace{\sin x}^{u}\, dx$$

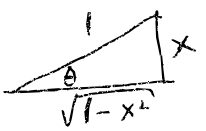
$$= e^x \cos x + e^x \sin x - \int e^x \cos x\, dx \quad I$$

$$2I = e^x(\sin x + \cos x)$$

$$I = \frac{e^x}{2}(\sin x + \cos x)$$

(Problem 5.43)

$$\int (\sin^{-1} x)^2\, dx$$



let $x = \sin\theta$

$dx = \cos\theta\, d\theta$

$$\int \underbrace{\theta^2}_{u} \underbrace{\cos\theta}_{dv/d\theta}\, d\theta$$

$$= \theta^2 \sin\theta - \int 2\theta \sin\theta\, d\theta$$

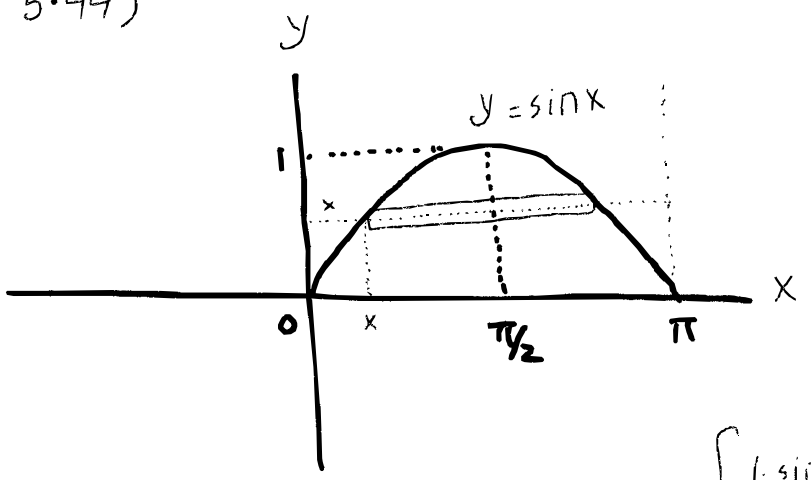
$$= \theta^2 \sin\theta - 2\left(\theta(-\cos\theta) - \int 1(-\cos\theta)\, d\theta\right)$$

$$= \theta^2 \sin\theta - 2(-\theta\cos\theta + \sin\theta + C)$$

$$= \theta^2 \sin\theta + 2\theta\cos\theta - 2\sin\theta + C$$

$$= x(\sin^{-1}(x))^2 + 2\sin^{-1}(x)\sqrt{1-x^2} - 2x + C$$

(Problem 5.44)



$R = \pi - \sin^{-1} y$

$r = \sin^{-1} y$

washer volume = (washer area) × dy

$= \pi(R^2 - r^2)\ dy$

$= \pi\left(\pi^2 - 2\pi(\sin^{-1}y) + (\sin^{-1}y)^2 - (\sin^{-1}y)^2\right) dy$

$= \pi^3 dy - 2\pi^2(\sin^{-1}y)\ dy$

$$\text{total volume} = \int_0^1 \pi^3\, dy - \int_0^1 2\pi^2 \sin^{-1}y\ dy$$

$$= \pi^3 - 2\pi^2 \int_0^1 \sin^{-1}y\ dy$$

$$= \pi^3 - 2\pi^2 \left[y\sin^{-1}y + \sqrt{1-y^2}\right]_0^1$$

$$= \pi^3 - 2\pi^2\left(1\cdot \pi/2 + 0 - 1\right) = \pi^3 - \pi^3 + 2\pi^2 \text{ unit}^3$$

$\int 1 \cdot \sin^{-1}y\ dy$

$dv/dx \quad u$

$= y\sin^{-1}y + \frac{1}{2}\int \frac{2}{\sqrt{1-y^2}}\, y\, dy$

$u = 1 - y^2$

$du = -2y\, dy$

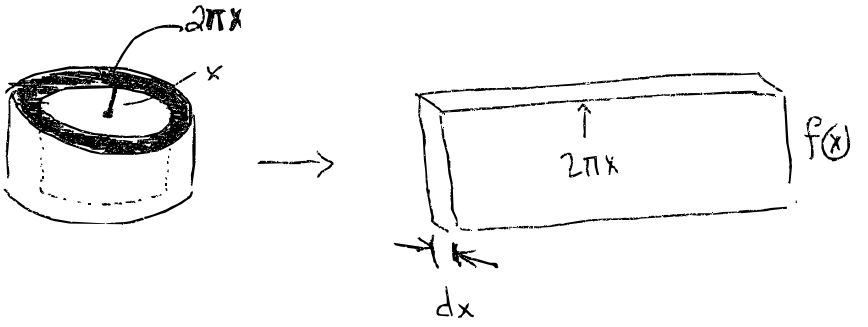
$= y\sin^{-1}y + \frac{1}{2}\int u^{-\frac{1}{2}} du$

$= y\sin^{-1}y + \frac{1}{2}\cdot\frac{1}{1/2}u^{1/2} + C$

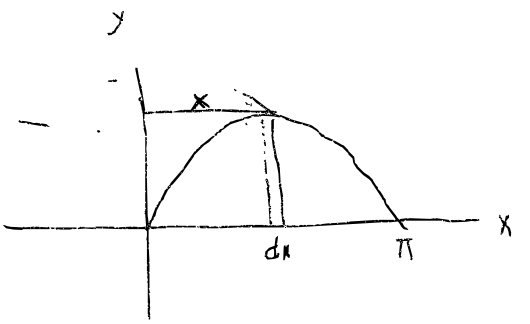
$= y\sin^{-1}y + \sqrt{1-y^2} + C$

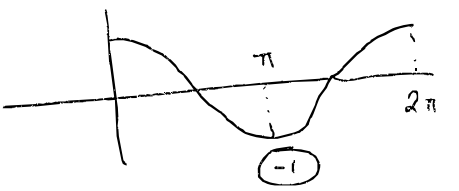
(Problem 5·45)

(a)



(c)





Volume of one slab $= (\sin x)(2\pi x)\,dx$

Total volume $= 2\pi \int_0^{\pi} \underset{u}{x} \underset{dv/dx}{\sin x}\, dx$

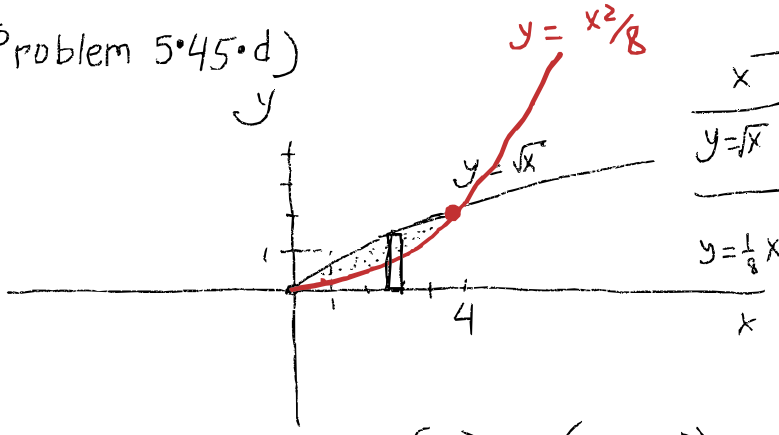
$= 2\pi \left[[x(-\cos x)]_0^{\pi} - \int_0^{\pi} 1 \cdot (-\cos x)\, dx \right]$

$= 2\pi \left[(\pi - 0) + \int_0^{\pi} \cos x\, dx \right]$

$= 2\pi \left[\pi + [\sin x]_0^{\pi} \right]$

$= \underline{2\pi^2}$ unit^3

(Problem 5·45·d)



x	1	4	9
$y=\sqrt{x}$	1	2	3
$y=\frac{1}{8}x^2$	$\frac{1}{8}$	2	16

slab's Volume $= (2\pi x)\,dx\left(\sqrt{x} - \frac{x^2}{8}\right)$

total volume $= 2\pi \int_0^4 \left(x^{3/2} - \frac{x^3}{8}\right) dx$

$$= 2\pi \left[\frac{2}{5}x^{5/2} - \frac{1}{8}\frac{1}{4}x^4\right]_0^4$$

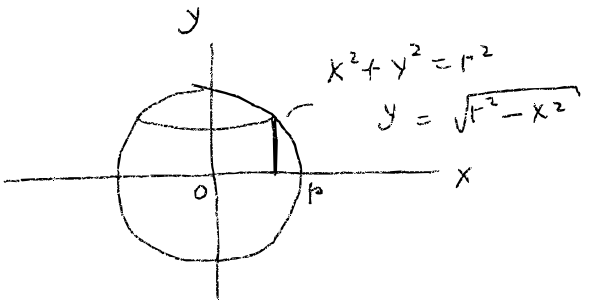
$$= 2\pi \left(\frac{2}{5}\cdot 32 - \frac{1}{32}4^4\right)$$

$$= 2\pi \left(\frac{2}{5}\cdot 32 - 8\right)$$

$$= 2\pi \cdot \frac{64-40}{5}$$

$$= \frac{48}{5}\pi \text{ unit}^3$$

(Problem 5.45.e)



slab's volume $= (2\pi x)\sqrt{r^2-x^2}\,dx$

Sphere's volume $= 2\int_0^r (2\pi x)\sqrt{r^2-x^2}\,dx$

$r^2-x^2 = u$

$-2x\,dx = du$

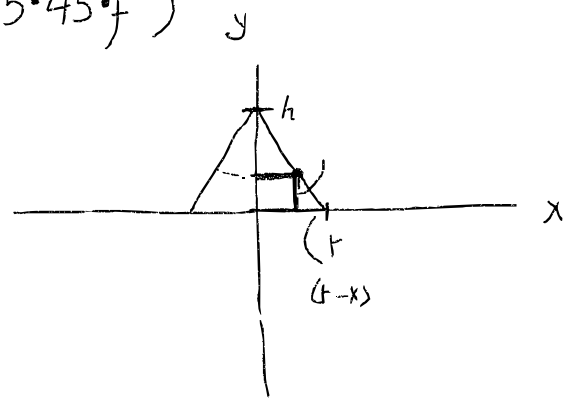
$$= -2\pi \int_{r^2}^{0} u^{1/2}\,du$$

$$= 2\pi \int_0^{r^2} u^{1/2}\,du$$

$$= 2\pi \cdot \frac{2}{3}\left[u^{3/2}\right]_0^{r^2}$$

$$= \frac{4}{3}\pi r^3$$ ∎

(Problem 5·45·f)



$(0,h)\ ;(r,0)$

$y - y_0 = m(x - x_0)$

$m = \frac{0-h}{r-0} = -\frac{h}{r}$

$y = -\frac{h}{r}(x-r)$

$y = h - \frac{h}{r}x$

$$\text{Volume} = \int_0^r 2\pi x\left(h - \frac{h}{r}x\right)dx$$

$$= 2\pi h\int_0^r x\,dx - \frac{2\pi h}{r}\int_0^r x^2\,dx$$

$$= 2\pi h \cdot \frac{1}{2} r^2 - \frac{2\pi h}{r}\cdot\frac{1}{3}r^3$$

$$= \pi r^2 h - \frac{2}{3}\pi r^2 h$$

$$= \frac{1}{3}\pi r^2 h \checkmark$$